CHM111: General Chemistry I: Exam 5
Dr. Deskins

$\frac{14}{15} \Rightarrow 93\%$

0	3	7	10	13	17	20	23	27	30	33	37	40	43	47	50	53	57	60	63	67	70	73	77	80	83	87	90	93	97	100
0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5	13	13.5	14	14.5	15

Name:

These questions assess your understanding of the material from Chapters 9 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Consider the reaction $2SO_3(g) \longrightarrow 2SO_2(g) + O_2(g)$. If 3.15 cm^3 of SO3 gas is completely consumed at the same temperature and pressure, what volume of O2 gas is produced? Justify your answer.

When $P_1 = P_2$ and $T_1 = T_2$, V is (only) dependent on molar ratios.

$3.15\,cm^3\ SO_3 \cdot \frac{1\ mol\ O_2}{2\ mol\ SO_3} = 1.575\,cm^3\ O_2$ produced

ANSWER: $1.58\,cm^3\ O_2$ produced
3SF ✓

2. A 13-L tank at 91.°C and 1.4 atm contains 7.2 g of nitrogen trifluoride (NF_3) and 9.37 g of bromine pentafluoride (BrF_5). Determine the mole fraction of bromine pentafluoride in the mixture. Justify your answer.

$P_1 = P_2$, $V_1 = V_2$, and $T_1 = T_2$

$\frac{0.0536}{0.0536 + 0.101} \approx 0.35$

$\frac{7.2\ g\ NF_3}{71\ g/mol} = 0.101\ mol\ NF_3$

$\frac{9.37\ g\ BrF_5}{174.9} = 0.0536\ mol\ BrF_5$

ANSWER: $0.35\ mol\ BrF_5/mol$ mixture
2SF ✓

3. The vapor pressure of a volatile solvent is measured at 75°C. The sample is then cooled to 35°C while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

Vapor pressure is proportional to temp., and decreasing temp. would decrease vapor pressure.

ANSWER: Decrease ✓

4. A sample of 11 g of sodium sulfide reacts completely with excess hydrochloric acid to produce sodium chloride and hydrogen sulfide gas in a sealed container. The container initially contains 0.19 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of hydrogen sulfide after the reaction is complete.

$Na_2S_{(s)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(s)} + H_2S_{(g)}$

$\frac{11\ g\ Na_2S}{78.1\ g/mol} = 0.141\ mol\ Na_2S \cdot \frac{1\ mol\ H_2S}{mol\ Na_2S} = 0.141\ mol\ H_2S$

$\frac{0.141}{0.141 + 0.19} \approx 0.426$ ✓

ANSWER: 0.43 ~~mol H_2S/mol mixture~~
2SF

5. A vertical cylinder has an internal diameter of 5.5-cm and contains gas beneath a movable piston. The gas column is 5.1-cm tall, and the gas pressure is measured to be 11.2 kPa. Determine the force exerted by the gas on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

Piston area $= 2.75^2\pi \approx 23.76\,cm^2$

$11.2\,kPa = \frac{11.2(1000)\,N}{m^2} \cdot \frac{1\,m^2}{1{,}000{,}000\,cm^2} = 0.0112\,N\,cm^{-2} \cdot 23.76 \approx 0.266\,N$

ANSWER: 0.27N
2SF

correct: $\boxed{27N}$

is $(100\,cm)^2 = (1\,m)^2$
⇓
1,000,000 ? ie is $100^2 = 10^6$?

1/2
16×
+4.5

6. A liquid is boiling at 32°C in a chamber where the external pressure is 160. kPa. Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Report the external pressure in SI units and justify your answer.

If a liquid is boiling $P_{Vapor} > P_A$

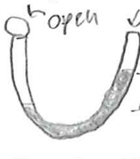
$160\,kPa \cdot \frac{1\,atm}{101.3\,kPa} = 1.579\,atm \approx 1.58\,atm$

ANSWER: Greater, 160 kPa = 1.58 atm

3SF

✓

7. An open-end mercury manometer is connected to a gas sample. The atmospheric ~~pressure is 739~~ torr, and the mercury is 45.1 mmHg higher on the end connected to the gas sample. Determine the gas pressure. Report your answer in SI units. Justify your answer.



open ✓closed

(height) higher

45.1 mmHg ~~lower~~ than outside

lower in P

739 torr − 45.1 torr = 693.9

ANSWER: 694

SF at ones place

units

1/2

$\frac{694\,mmHg}{} \cdot \frac{101325\,Pa}{760\,torr \text{ or } mmHg}$

or 694 torr

9.25 × 10⁴ Pa

correct #7

8. A high-pressure gas cylinder is allowed to fill a large empty collection bag. The cylinder initially contains gas at 3.2 atm. The valve is opened until the pressure inside the cylinder becomes atmospheric pressure and is then closed. Determine the fraction of the original gas that remains inside the cylinder. Justify your answer.

$P_1V_1 = P_2V_2$ $\qquad \frac{P_2V_2}{P_1} = V_1$

$\frac{V_1}{V_2} = \frac{P_2}{P_1}$ $\qquad \frac{P_2}{P_1} = \frac{1}{3.2} = 0.3125 \approx 0.31$

ANSWER: 0.31 units gas after / before (or 31%)

2SF

✓

9. A flexible latex balloon contains 2.80 L of argon at −20.6°C while indoors. It is then taken outdoors, where the air temperature is 22°C. Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ $\qquad V_2 = \frac{V_1T_2}{T_1} = \frac{2.8 \cdot 295.15}{252.55} \approx 3.27\,L$

ANSWER: 3.3 L

2SF

✓

10. A 858-mL sample of hydrogen fluoride (HF) is initially at 489 K. Its boiling point is 20.°C (293.15 K). The pressure is then reduced to 54% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest °C. Justify your answer.

ANSWER: No, condenses

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\qquad T_2 = \frac{P_2T_1}{P_1} = \frac{0.54(489)}{1} = 264.06\,°K < 293.15$

✓

11. An unknown gas has a mass of 36.1 g and occupies 13 L at 86.6 kPa and 23.°C. Determine the molar mass of the gas. Justify your answer. Report your answer in SI units.

$PV = nRT$ $\qquad n = \frac{m}{M}$ $\qquad PV = \frac{mRT}{M}$

ANSWER: 79 g/mol

2SF

✓

$M = \frac{mRT}{PV}$ $\qquad \frac{36.1(0.08206)(296.15)}{0.855(13)} \approx 78.93\,g/mol$

This number makes sense because $\frac{78.93}{36.1} \approx \frac{22.4}{13}$

$86.6\,kpa \cdot \frac{1\,atm}{101.3\,kPa} = 0.85489\,atm$

+5.5

12. Elizabeth is preparing a deep-sea submersible. A rigid cylindrical air tank has an internal diameter of 33.0-cm and an internal height of 116-cm. After filling, the pressure inside the tank is 210. atm and the temperature is 296 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

Cylindrical area $= r^2 \pi h = (16.5^2)(116)(\pi)$

Area $\approx$ 99.2 ~~[illegible]~~ $cm^3 \cdot \frac{1 L}{1000 cm^3} = 99.21 L$

$PV = nRT$ $\quad n = \frac{PV}{RT}$ $\quad n = \frac{210 \cdot 99.21}{0.08206 \cdot 296} = 857.7$ moles air.

ANSWER: 858 moles air

3SF

13. A sample reported to be HBr has a mass of 50. g and occupies 11 L at 140 kPa and 24.°C. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

$M = \frac{mRT}{PV}$ $\quad M = \frac{50 \cdot 0.08206 \cdot 297.15}{11 \cdot 1.382}$

$M = 80.2$ (g/mol) $\quad 80.2 \approx 81$ g/mol

$M_{HBr} \approx 81$ (g/mol) $\quad$ g/mol

ANSWER: Slightly impure HBr or small measurement errors and pure HBr.

good

14. The reaction $2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(g)$ occurs with all gases measured at the same temperature and pressure. If 6.0 cm^3 of carbon dioxide is consumed, what volume of ethane is produced or consumed? Justify your answer.

~~[illegible]~~ Under constant T & P, V is only dependent on n.

$4CO_2 \rightarrow 2C_2H_6$ $\quad 6.0 cm^3 CO_2 \cdot \frac{2 mol C_2H_6}{4 mol CO_2} = 3.0 cm^3 C_2H_6$

ANSWER: 3.0 cm^3 C_2H_6 produced

2 SF

15. A sample of an unknown gas is transferred into a dry gas bulb. The sample has a mass of 1.660 g, occupies 0.500 L at 100.0 kPa and 293.15 K. Possible identities are: CO, H_2O, HBr, CO_2, Cl_2, UF_6, Kr, SO_2, CH_4, Xe, NO, F_2, C_2H_2, Ne, H_2S, Ar, SF_6, He, H_2. Determine the identity of the unknown gas from the given choices. Justify your answer.

$M = \frac{mRT}{PV}$ $\quad M = \frac{1.66 \cdot 0.08206 \cdot 293.15}{0.987 \cdot 0.5}$

$M_{sample} = 80.917 \approx M_{HBr} = 80.9$ g/mol

g/mol

ANSWER: HBr

+4

1. $1.58\,cm^3$. SINCE at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:1 ratio of SO3 to O2, so the product volume is multiplied by 1/2.
2. 0.35. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
3. Decreases. ∵ lowering the temperature reduces the kinetic energy of the liquid molecules, so fewer molecules have sufficient energy to enter the vapor phase and the equilibrium vapor pressure decreases.
4. 0.43. ∵ 11 g of sodium sulfide corresponds to 0.14 mol. The reaction produces 0.14 mol of hydrogen sulfide. The amount of neon remains 0.19 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
5. 27 N Only the piston area is needed because pressure is force per unit area; the gas height does not affect the force once the pressure is known. ∵ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
6. 1.58 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
7. $9.25 \times 10^4\,Pa$ ∵ the mercury is higher on the gas-sample side, the gas pressure is lower than atmospheric pressure, and the pressure difference is subtracted from the atmospheric pressure before converting to SI units.
8. 0.31. ∵ the cylinder volume does not change, reducing the pressure from the initial value to atmospheric pressure reduces the amount of gas in direct proportion. The remaining fraction equals the final pressure divided by the initial pressure.
9. 3.3 L. ∵ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
10. No. ∵ keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
11. 79 g/mol. ∵ the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
12. 858 mol. ∵ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law, $PV = nRT$. A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for n.
13. Consistent. ∵ The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which matches the molar mass of HBr.
14. $3.0\,cm^3$. At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the CO2:C2H6 coefficient ratio of 0.5:1 determines the required volume.
15. HBr. ∵ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 80.91 g/mol. Only HBr has that molar mass, so it is the only identity consistent with the measurements.